Differentiation & Integration

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Differentiation, Rules & Rates of Change · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Integration, Areas & Kinematics · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A curve is defined parametrically by $x = t^2 + 1$, $y = 2t^3 - 3t$. Find $\frac{dy}{dx}$ in terms of t and the value at $t = 1$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6t^2 - 3$. $\frac{dy}{dx} = \frac{6t^2 - 3}{2t}$. At $t = 1$: $\frac{3}{2}$.

6 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 3.01 Graphs & Transformations**

[Logarithms, Exponentials & Straight-Line Reduction · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 4.01 Coordinate Geometry**

[AI SL — Specific Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 5.01 Trigonometry**

[Overview · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 7.01 Statistics & Probability**

[Number & Algebra · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Functions: Composite & Inverse

[Calculus · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Differentiate $y = 3x^4 - 5x^2 + 7x - 2$ with respect to x.

a. $12x^3 - 10x + 7$ ✓

b. $12x^3 - 10x^2 + 7$

c. $7x^3 - 10x + 7$

d. $12x^4 - 10x^2 + 7$

Apply the power rule: $d/dx[3x^4] = 12x^3$, $d/dx[-5x^2] = -10x$, $d/dx[7x] = 7$, $d/dx[-2] = 0$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the gradient of the curve $y = x^3 - 4x + 1$ at the point where $x = 2$.

a. **8** ✓

b. 12

c. 4

d. 0

$dy/dx = 3x^2 - 4$. At $x = 2$: $dy/dx = 3(4) - 4 = 12 - 4 = 8$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$.

a. $y = 3x - 7$ ✓

b. $y = 3x - 2$

c. $y = x - 1$

d. $y = 2x - 4$

$dy/dx = 2x - 3$. At $x = 3$, gradient $m = 2(3) - 3 = 3$. Tangent: $y - 2 = 3(x - 3) \Rightarrow y = 3x - 7$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the stationary points of the curve $y = 2x^3 - 9x^2 + 12x + 1$ and determine their nature.

a. **Local max at $(1, 6)$, local min at $(2, 5)$** ✓

b. Local min at $(1, 6)$, local max at $(2, 5)$

c. Local max at $(2, 5)$, inflection at $(1, 6)$

d. Local min at $(0, 1)$, local max at $(2, 5)$

$dy/dx = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) = 0 \Rightarrow x = 1, 2$. $d^2y/dx^2 = 12x - 18$. At $x = 1$: $d^2y/dx^2 = -6 < 0$ (max, $y = 6$). At $x = 2$: $d^2y/dx^2 = 6 > 0$ (min, $y = 5$).

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A metal wire of length 40 cm is bent into a rectangle of dimensions x by y. What value of x maximizes the enclosed area?

a. $x = 10$ cm ✓

b. $x = 20$ cm

c. $x = 5$ cm

d. $x = 15$ cm

$2x + 2y = 40 \Rightarrow y = 20 - x$. Area $A = x(20 - x) = 20x - x^2$. $dA/dx = 20 - 2x = 0 \Rightarrow x = 10$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what range of values of x is the function $f(x) = x^3 - 6x^2 + 9x - 4$ strictly increasing?

a. $x < 1$ **or** $x > 3$ ✓

b. $1 < x < 3$

c. $x > 2$

d. $x < 2$

$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) > 0 \Rightarrow x < 1$ or $x > 3$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate the definite integral $\int_1^3 (3x^2 - 2x)\,dx$.

a. **18** ✓

b. 24

c. 16

d. 20

$[x^3 - x^2]_1^3 = (27 - 9) - (1 - 1) = 18 - 0 = 18$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A curve passes through $(1, 4)$ and has gradient function $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$ ✓

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 - x + 1$

$y = \int (4x - 1)dx = 2x^2 - x + C$. At $(1, 4)$: $4 = 2(1) - 1 + C \Rightarrow 4 = 1 + C \Rightarrow C = 3$. Curve: $y = 2x^2 - x + 3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. $\frac{64}{3}$

Roots at $x = \pm 2$. Area $= \int_{-2}^{2} (4 - x^2)dx = [4x - x^3/3]_{-2}^{2} = (8 - 8/3) - (-8 + 8/3) = 16/3 - (-16/3) = 32/3$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$ ✓

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

Intersections: $x^2 = 2x \Rightarrow x(x - 2) = 0 \Rightarrow x = 0, 2$. Area $= \int_0^2 (2x - x^2)dx = [x^2 - x^3/3]_0^2 = 4 - 8/3 = 4/3$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A particle moves along a straight line such that its velocity at time t is $v(t) = 3t^2 - 12t + 9$. Find the total displacement between $t = 0$ and $t = 3$.

a. **0** ✓

b. 9

c. 18

d. 4.5

Displacement $= \int_0^3 (3t^2 - 12t + 9)dt = [t^3 - 6t^2 + 9t]_0^3 = (27 - 54 + 27) - 0 = 0$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.

a. $\frac{32}{3}$ ✓

b. 16

c. $\frac{64}{3}$

d. $\frac{16}{3}$

To find the area enclosed by the curve and the x-axis, first find the x-intercepts by setting $y = 0$: $4x - x^2 = 0 \Rightarrow x(4 - x) = 0 \Rightarrow x = 0$ or $x = 4$. The area is given by the definite integral of y with respect to x from $x = 0$ to $x = 4$: $\int_0^4 (4x - x^2)\,dx = \left[2x^2 - \frac{x^3}{3}\right]_0^4$ Evaluating between the limits: $\left(2(4)^2 - \frac{4^3}{3}\right) - 0 = 32 - \frac{64}{3} = \frac{96 - 64}{3} = \frac{32}{3}$. Common misconceptio.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.

a. $y = 2x + 1$

b. $y = 6x - 1$

c. $y = 6x + 1$ ✓

d. $y = 6x + 3$

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(2x + 1)^2 \cdot (2) = 6(2x + 1)^2$. At $x = 0$, slope $m = 6$. Equation: $y - 1 = 6(x - 0) \Rightarrow y = 6x + 1$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.

a. $(1, 6)$**, which is a local minimum** ✓

b. $(1, 6)$, which is a local maximum

c. $(-1, -2)$, which is a local minimum

d. $(2, 10)$, which is a local minimum

1. **Find the point of tangency:** Substitute $x = 2$ into the tangent line equation $y = 7x - 4$: $y = 7(2) - 4 = 10$ Since the line touches the curve at $x = 2$, the point $(2, 10)$ lies on the curve $y = ax^2 + \frac{b}{x}$: $a(2)^2 + \frac{b}{2} = 10 \Rightarrow 4a + \frac{b}{2} = 10 \Rightarrow 8a + b = 20 \quad --- \ (1)$ 2. **Use the gradient of the tangent:** Differentiating $y = ax^2 + bx^{-1}$ gives: $\frac{dy}{dx} = 2ax - \frac{b}{x^2}$ The gradient of the tangent line $y = 7x - 4$ is 7. At.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2. During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm.

a. 20π ✓

b. 50π

c. $\frac{36\pi}{5}$

d. 11π

1. **Express the surface area constraint:** The total exterior surface area consists of the flat circular base, the curved cylindrical wall, and the hemispherical dome: $S = \pi r^2 + 2\pi rh + 2\pi r^2 = 3\pi r^2 + 2\pi rh$ Given $S = 180\pi$: $3\pi r^2 + 2\pi rh = 180\pi \Rightarrow 3r^2 + 2rh = 180$ 2. **Find the value of r when $h = 16.5$ cm:** $3r^2 + 2(16.5)r = 180 \Rightarrow 3r^2 + 33r - 180 = 0$ Dividing by 3: $r^2 + 11r - 60 = 0 \Rightarrow (r + 15)(r - 4) = 0$ Since $r > 0$, we have.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$ ✓

b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

1. **Find the stationary point M:** Using the quotient rule to differentiate $y = \frac{\ln x}{x^2}$: $\frac{dy}{dx} = \frac{\frac{1}{x} \cdot x^2 - (\ln x)(2x)}{x^4} = \frac{x(1 - 2\ln x)}{x^4} = \frac{1 - 2\ln x}{x^3}$. Setting $\frac{dy}{dx} = 0$: $1 - 2\ln x = 0 \Rightarrow \ln x = \frac{1}{2} \Rightarrow x = e^{1/2} = \sqrt{e}$. The y -coordinate at M is: $y = \frac{\ln(\sqrt{e})}{(\sqrt{e})^2} = \frac{\frac{1}{2}}{e} = \frac{1}{2e}$. Since M is a stationary point, the tangent line L is horizontal wit.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The variables x and y are connected by the equation $y = \sqrt{2x + 5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

a. 2 units per second ✓

b. 1 unit per second

c. $\frac{1}{3}$ units per second

d. 12 units per second

We are given $y = (2x + 5)^{\frac{1}{2}}$ and $\frac{dx}{dt} = 6$. Step 1: Differentiate y with respect to x using the chain rule: $\frac{dy}{dx} = \frac{1}{2}(2x + 5)^{-\frac{1}{2}} \cdot \frac{d}{dx}(2x + 5) = \frac{1}{2\sqrt{2x+5}} \cdot 2 = \frac{1}{\sqrt{2x+5}}$ Step 2: Evaluate $\frac{dy}{dx}$ at $x = 2$: $\frac{dy}{dx}\Big|_{x=2} = \frac{1}{\sqrt{2(2)+5}} = \frac{1}{\sqrt{9}} = \frac{1}{3}$ Step 3: Use.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry

The curve C has equation $y = \frac{ax + 6}{x^2 + 3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$ ✓

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

To find the unknown parameter a, differentiate y with respect to x using the quotient rule: $\frac{dy}{dx} = \frac{a(x^2 + 3) - (ax + 6)(2x)}{(x^2 + 3)^2} = \frac{3a - 12x - ax^2}{(x^2 + 3)^2}$ At a stationary point, $\frac{dy}{dx} = 0$, which requires the numerator to be zero: $3a - 12x - ax^2 = 0$ Given that $x = 1$ is a stationary point:

$3a - 12(1) - a(1)^2 = 0 \Rightarrow 2a - 12 = 0 \Rightarrow a = 6$ Substitute $a = 6$ back into the numerator of the derivative.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.

a. 2 ✓

b. 4

c. 3

d. 1

Step 1: Find the points of intersection of the curve and the line. Set the equations equal to each other: $6x - x^2 = mx \Rightarrow x^2 - (6 - m)x = 0 \Rightarrow x(x - (6 - m)) = 0$ Since the line intersects the curve at the origin $(0, 0)$ and at point P, the x-coordinate of P is $x = 6 - m$. Step 2: Set up the definite integral for the enclosed area. For $0 \le x \le 6 - m$, the curve lies above the straight line. The enclosed area A is given by.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2 ✓

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate y with respect to x using the chain rule. Let $u = 2x^3 + 1$, so $y = \ln(u)$. Then: $\frac{du}{dx} = 6x^2$ and $\frac{dy}{du} = \frac{1}{u}$ Applying the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3 + 1}$ Now evaluate the derivative at $x = 1$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Circular Measure

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$ ✓

b. $6e^3$

c. $3e^3$

d. $2e^3$

To find the gradient of the curve, differentiate $y = x^2e^{3x}$ with respect to x using the product rule: $\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$ Let $u = x^2$ and $v = e^{3x}$. Then $\frac{du}{dx} = 2x$ and, by the chain rule, $\frac{dv}{dx} = 3e^{3x}$. Applying the product rule: $\frac{dy}{dx} = x^2(3e^{3x}) + 2x(e^{3x}) = (3x^2 + 2x)e^{3x}$ Substitute $x = 1$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.

a. 57 units per second ✓

b. 19 units per second

c. $\frac{19}{3}$ units per second

d. 21 units per second

To find the rate of change of y with respect to time, apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ 1. Differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ 2. Evaluate $\frac{dy}{dx}$ at $x = 2$: $\frac{dy}{dx}\Big|_{x=2} = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = 19$ 3. Multiply by.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Series & Binomial Theorem

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.

a. 76 units s^{-1} ✓

b. 19 units s^{-1}

c. 28 units s^{-1}

d. 24 units s^{-1}

To find the rate of change of y with respect to time t, we apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ First, differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ Substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 6(2)^2 - 5 = 24 - 5 = 19$ Given that.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Kinematics

A curve has equation $y = \frac{kx}{2x-1}$, where $xeq\frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.

a. -0.2 units per second ✓

b. 0.2 units per second

c. -1.8 units per second

d. -0.6 units per second

Step 1: Find the derivative $\frac{dy}{dx}$ in terms of k using the quotient rule. Let $u = kx$ and $v = 2x - 1$. Then $u' = k$ and $v' = 2$. $\frac{dy}{dx} = \frac{k(2x-1) - 2(kx)}{(2x-1)^2} = \frac{2kx - k - 2kx}{(2x-1)^2} = \frac{-k}{(2x-1)^2}$ Step 2: Find the parameter k. At $x = 1$, the gradient of the tangent is: $m_T = \frac{dy}{dx}\Big|_{x=1} = \frac{-k}{(2(1)-1)^2} = -k$ The gradient of the normal is given by $m_N = -\frac{1}{m_T} = \frac{1}{k}$. Since the gradient of the normal is.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Forces & Equilibrium

Find the exact value of the definite integral $\int_0^{\ln 2} e^{2x}\,dx$.

a. $\frac{3}{2}$ ✓

b. 3

c. $\frac{1}{2}$

d. $\frac{7}{2}$

To evaluate the definite integral: 1. Find the antiderivative using the standard standard result $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C$: $\int e^{2x}\,dx = \frac{1}{2}e^{2x}$ 2. Apply the limits from $x = 0$ to $x = \ln 2$: $\left[\frac{1}{2}e^{2x}\right]_0^{\ln 2} = \frac{1}{2}e^{2\ln 2} - \frac{1}{2}e^{2(0)}$ 3. Simplify the exponential terms using the power law of logarithms, $2\ln 2 = \ln(2^2) = \ln 4$.

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